

M5 + M6 Student Notes — ETL Options and Automation

Notes for the compressed Day 1 block (4:05–4:20 PM CT). M5 surveys the tools for "when SQL is not enough" — conceptual only, no lab. M6 covers automation techniques: the block that maps most directly to your SQL Agent jobs and CI/CD world.

M5 — when SQL is not enough (conceptual)

- **Default to ELT in BigQuery.** Reach for M5 tools only when transformation needs more than SQL — custom code, streaming latency, or non-BigQuery destinations.
- **Batch vs streaming is a latency decision, not a religion.** Nightly or micro-batch loads cover most warehouse work; streaming is for when the business genuinely needs seconds-to-minutes freshness.
- **Pub/Sub** is the managed messaging backbone: producers publish events, subscribers consume them — the durable, serverless landing pad for anything streaming into GCP.
- **Dataflow** is the managed runner for Apache Beam pipelines (batch and stream), and **templates** are prebuilt pipelines you configure rather than code (for example, Pub/Sub → BigQuery) — common patterns without Beam development.
- **Depth lives elsewhere:** Spark and the streaming material are later-course topics (Course 4); the M5 Spark lab is optional homework. Today's goal is knowing the names and when to reach for each.

Who on your team cares — M5: Developers, if ingestion ever goes real-time (Pub/Sub and Dataflow become your problem then). **M&O**, awareness level only — these are the boxes that will show up on future architecture diagrams.

M6 — automation: your CI/CD analogue

- **Cloud Scheduler ≈ SQL Agent.** Managed cron that triggers targets (HTTP endpoints, Pub/Sub messages). The general-purpose timer — no scheduler VM to patch.
- **Scheduled queries** (the 4:20 PM closing demo) are the simplest case of all: one SQL statement on a schedule, overwriting a destination table. A SQL Agent job reduced to its essence.
- **Dataform release configurations + scheduled workflow invocations** put your Silver→Gold runs on a cadence, compiled from a git commit — the lab you just finished, productionalized. This is your deployment discipline applied to SQL in git.
- **Cloud Composer** is managed Apache Airflow: the orchestrator for when you have many interdependent pipelines — cross-system dependencies, retries, SLAs. More than you need on day one; the answer when "run A, then B, then C across three systems" gets complicated.
- **Eventarc** routes events (a file lands in a bucket, a log entry appears) to targets — the trigger half of event-driven loads. **Cloud Run functions** execute small pieces of code in response — the action half.

Who on your team cares — M6: M&O, this is your block. Scheduling, orchestration, and failure visibility are the operating model for the new platform. **Developers** care mainly about how their Dataform releases get triggered — a scheduled invocation, Composer, or your existing CI/CD calling the API.

Choosing an automation tool

Service	What it is	Reach for it when	Your analogue
Scheduled queries	A saved query on a schedule	The whole job is one SQL statement	Single-step SQL Agent job
Cloud Scheduler	Managed cron triggering HTTP/Pub/Sub targets	"Every day at 6 AM, start something"	SQL Agent schedule
Dataform release + workflow schedule	Compile-from-git plus scheduled execution	Running your Silver→Gold workflow on a cadence	Deployment pipeline + job
Cloud Composer	Managed Apache Airflow (DAGs)	Many interdependent, cross-system pipelines	Master package / enterprise scheduler
Eventarc	Event routing between services	"When X happens, kick off Y"	File-watcher / event-driven triggers
Cloud Run functions	Small event-triggered code	Glue logic too small for a pipeline	Script Task / custom .NET glue

Key takeaways

- Pick by shape, not by fashion: one statement → **scheduled query**; your transform flow → **Dataform release config**; arbitrary things on a timer → **Cloud Scheduler**; many pipelines with dependencies → **Composer**; events instead of clocks → **Eventarc** ♦ **Cloud Run functions**.
- Automation in GCP is serverless: no scheduler box to maintain, no agent service — you declare the schedule or the trigger, Google runs it.
- Nothing here changes your discipline — source control, review, staged deployment, monitoring. It changes where the buttons are.
- The M5 tools are not a detour from your medallion plan; they are the escape hatches you will reach for at the edges (streaming sources, non-SQL transforms) once the core is running.

Your SQL Server world → GCP world

In your current stack	In GCP
SQL Agent job (single T-SQL step)	Scheduled query
SQL Agent job schedule	Cloud Scheduler
Deployment pipeline for SQL changes	Dataform release configuration + workflow schedule
Master SSIS package orchestrating many packages	Cloud Composer (managed Airflow)
File/event watchers kicking off jobs	Eventarc
Custom .NET / Script Task glue	Cloud Run functions
Service Broker / queue-based decoupling	Pub/Sub

Dig deeper

- [Cloud Scheduler documentation](#)
- [Cloud Composer \(managed Apache Airflow\)](#)
- [Pub/Sub overview](#)
- [Dataflow overview](#)
- [Eventarc overview](#)
- [Cloud Run functions overview](#)
- [Scheduling queries in BigQuery](#)

Related: the [closing activity guide](#) — schedule your first query tonight, in five minutes.

Part of the Iowa DOM DoIT Data Engineering student pack — see ../00_schedule.md.